

Lec-10 : Intro to Neural Networks

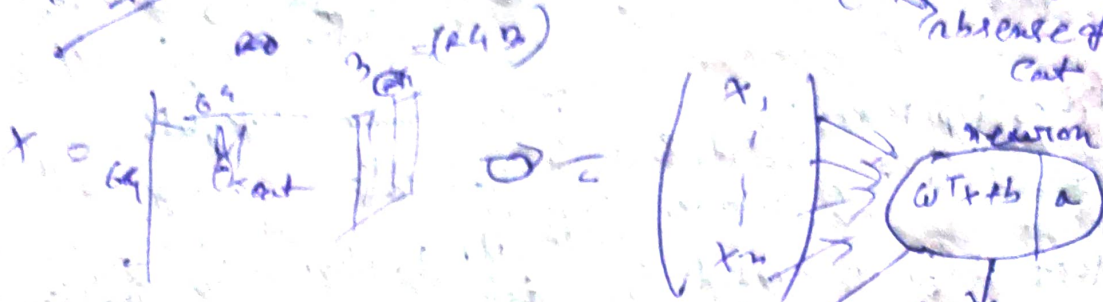
Contents :
 Logistic Regression
 Neural Networks

Deep learning : (used in vision, NLP, Text classification)
 Properties:

- 1) Computational power ($\uparrow$)
- 2) Data availability ($\uparrow$)
- 3) Algorithming.

Logistic Regression:

Goal of goal $\rightarrow$ Find cats in images $\begin{cases} 1 \rightarrow \text{presence of cat} \\ 0 \rightarrow \text{absence of cat} \end{cases}$



The process of training:

(i) Initialize $w, b \rightarrow$ 2.5ms. (weights)

(ii) Find the optimal w, b

(iii) use $\hat{y} = \sigma(wx + b)$ to predict

for Linear Regression

Total parameters here = $(12,288 + 1)$
 $\downarrow \quad \downarrow$
 $w \quad b$
 $= (12,289)$

(ReLU) for linear regression & sigmoid for logistic regression
 $\hat{y} = \sigma(wx + b)$
 $\hat{y} = \sigma(1, 12,288) = (12,288, 1)$

Logistic loss function
 $J = -y \log \hat{y} - (1-y) \log(1-\hat{y})$
 Minimize

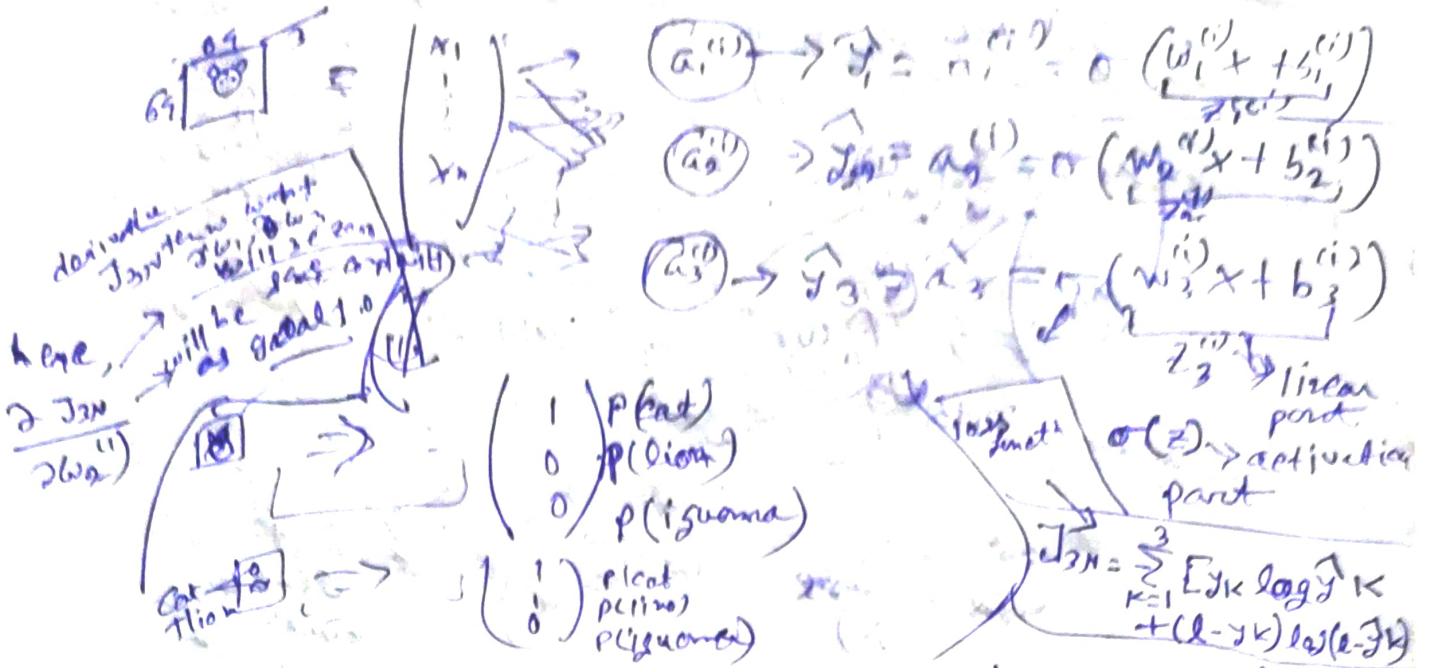
$w = w - 2 \frac{\partial J}{\partial w}$
 $b = b - 2 \frac{\partial J}{\partial b}$

Two eqns needed $\Rightarrow$

Eq 1 : neuron = linear + activation

Eq 2 : model = architecture + parameters
 (a model in industry has 2 files $\rightarrow$ architecture + parameters)

Goal 2.0 : find cat/lion/iguana in images



(let say we want to predict ~~any~~ just 1 to classify it and if that label is very rare (rare disease) so, N.N model will not be the perfect match for them becz the given data has also has the disease rate, so, they won't notice that)

to solve this :- use probability distribution

in the o/p layer $\rightarrow$ as we in prob distrib all o/p probabilities depends upon each other and they sum up to one $\Rightarrow$ the way if there is no cat and lion in data then N.N gives most prob dist to (iguana).

Goal 2.0 $\Downarrow$
Software does this $\Rightarrow$

$$\begin{aligned}
 \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} &= \begin{pmatrix} z_1^{(1)} \\ z_2^{(1)} \\ z_3^{(1)} \end{pmatrix} \\
 &= \frac{e^{z_1^{(1)}}}{\sum_{k=1}^3 e^{z_k^{(1)}}} \\
 &= \frac{e^{z_2^{(1)}}}{\sum_{k=1}^3 e^{z_k^{(1)}}} \\
 &= \frac{e^{z_3^{(1)}}}{\sum_{k=1}^3 e^{z_k^{(1)}}}
 \end{aligned}$$

cat lion iguana

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

loss function $\rightarrow$ cross-entropy loss

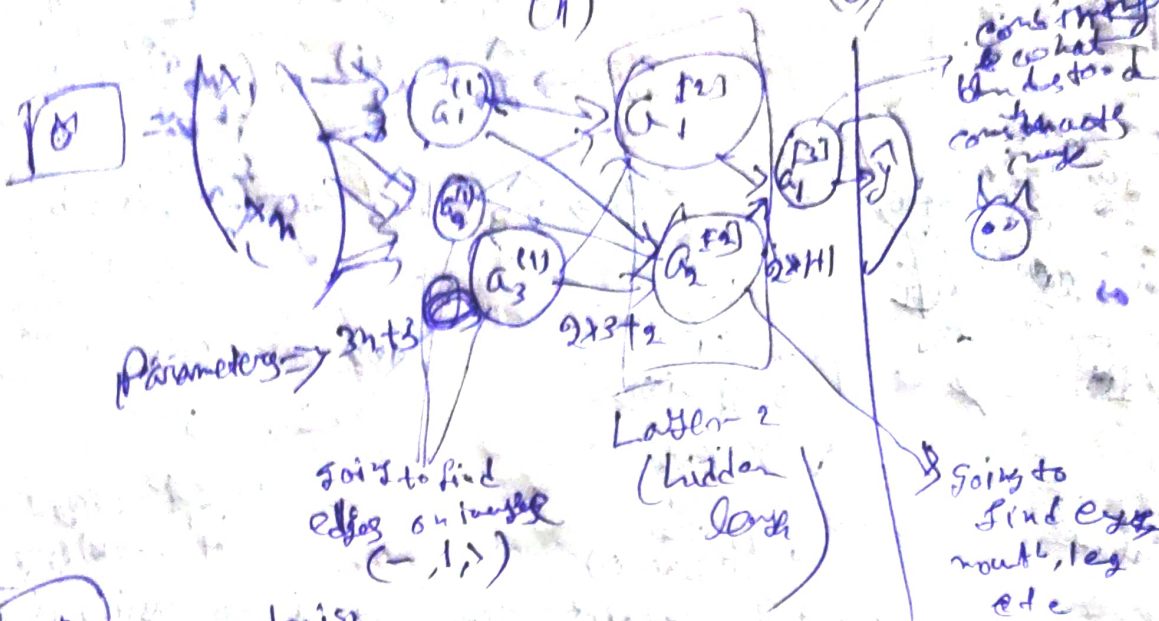
Software multiclass regression network

$$J(F) = \sum_{k=1}^3 y_k \log \hat{y}_k + (1-y_k) \log (1-\hat{y}_k)$$

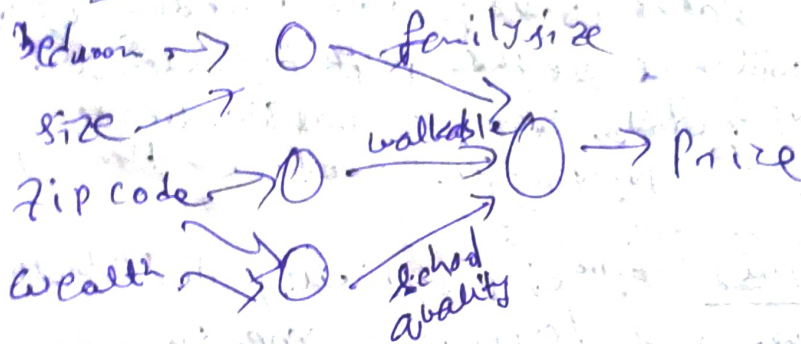
$w_2 = 1.5$

II. Neural Networks →

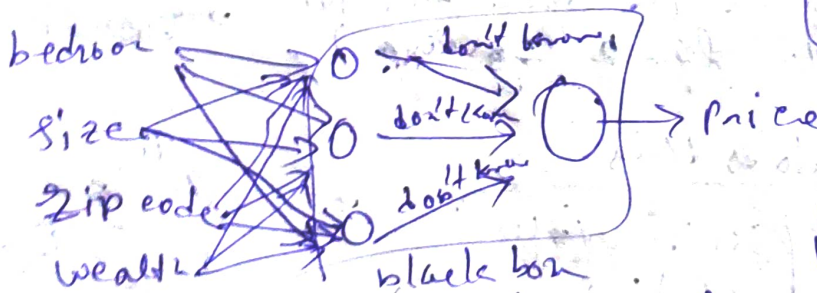
Goal → Image → cat (1) v/s no cat (0)



Q-2 Human design



Actual fully connected network



(End to end learning / black box model)

How to know how many neurons should be there in any layer / what should be the architecture?

By Training 10 diff architectures
Looking validation set (accuracy, etc) → decide which is best
Predicting based on test set

The complex the problem → the more data should be given & the deeper the network becomes

cat classification v/s no cat
→ random guess

Propagation Equations :-

(Shape of these are in vector)

$$(3,1) \rightarrow Z^{[1]} = W^{[0]} \cdot x^{[0]} + b^{[1]} \quad (3,1)$$

$$(3,1) \rightarrow a^{[1]} = \sigma(Z^{[1]})$$

$$(2,1) \rightarrow Z^{[2]} = W^{[1]} \cdot a^{[1]} + b^{[2]} \quad (2,1)$$

$(2,1) \rightarrow a^{[2]} = 0 \cdot (z^{[2]})$
 $(1,1) \rightarrow z^{[3]} = w^{[3]} \cdot (1,2) a^{[2]} + b^{[3]} \cdot (1,1) w^{[1]}$
 $y = a^{[3]} = 0 \cdot (z^{[3]}) \cdot (2,1)$
 $(1,1)$

S. G. D

updates parameters
for each examples
making the update
noisy, fluctuating
with ~~new~~ every other
new ~~noisy~~ ~~the~~ examples

Order C/D

updates parameters after seeing whole bunch of parameters, making the update precise to the point.

- What happens for an input batch of m examples?

$X =$ 

$$Z = \overbrace{W^{(3,m)}}^{(3,m)} \leftarrow (m,m) + 6 \overbrace{W^{(3,1)}}^{(3,1)}$$

$(3, m) = \begin{bmatrix} z^{(0)}(1) & z^{(0)}(m) \\ z^{(1)}(1) & z^{(1)}(m) \\ \vdots & \vdots \\ z^{(n-1)}(1) & z^{(n-1)}(m) \end{bmatrix}$

To solve this
shape is inscribed
we use

"Broadcasting" →

In many case directly do the broadcasting

[illegible]

④ Optimization Problem $\Rightarrow$

Optimizing: $w^{[1]}, w^{[2]}, w^{[3]}, b^{[1]}, b^{[2]}, b^{[3]}$

Define Loss / Cost function

(group) (example)

$$J(\gamma, z) = \frac{1}{n} \sum_{i=1}^n L(i)$$

with $L(i) = - \left[y^{(i)} \log \gamma^{(i)} + (1 - y^{(i)}) \log (1 - \gamma^{(i)}) \right]$

$\nwarrow$
logistic loss

Completing $\Rightarrow$

$$\frac{\partial J}{\partial w^{[2]}} \text{ v/s } \frac{\partial J}{\partial w^{[3]}}$$

It doesn't matter for derivation we will go with $\frac{\partial J}{\partial w^{[2]}}$ starting from inside out

④ Backward Propagation $\Rightarrow$

$$\forall l = 1 \dots 3 \quad w^{[l]} = w^{[l]} - \alpha \frac{\partial J}{\partial w^{[l]}}$$

$\uparrow$
layers

$$b^{[l]} = b^{[l]} - \alpha \frac{\partial J}{\partial b^{[l]}}$$

derivation should go backwards. be computing

$\frac{\partial J}{\partial w^{[2]}}$ will make the computations of $\frac{\partial J}{\partial w^{[2]}}$, $\frac{\partial J}{\partial w^{[3]}}$

log. In short back chain rule.

$$\frac{\partial J}{\partial w^{[3]}} = \frac{\partial J}{\partial z^{[3]}} \cdot \frac{\partial z^{[3]}}{\partial w^{[3]}} = \frac{\partial J}{\partial a^{[3]}} \cdot \frac{\partial a^{[3]}}{\partial z^{[3]}} \cdot \frac{\partial z^{[3]}}{\partial w^{[3]}}$$

$$\frac{\partial J}{\partial w^{[2]}} = \left(\frac{\partial J}{\partial z^{[3]}} \right) \cdot \frac{\partial z^{[3]}}{\partial a^{[2]}} \cdot \frac{\partial a^{[2]}}{\partial z^{[2]}} \cdot \frac{\partial z^{[2]}}{\partial w^{[2]}}$$

$$\frac{\partial J}{\partial w^{[1]}} = \left(\frac{\partial J}{\partial z^{[2]}} \right) \cdot \frac{\partial z^{[2]}}{\partial a^{[1]}} \cdot \frac{\partial a^{[1]}}{\partial z^{[1]}} \cdot \frac{\partial z^{[1]}}{\partial w^{[1]}}$$

(Talking various things in NN will make N-N alive)